

1.1 Introduction

The LPC176x/5x is an ARM Cortex-M3 based microcontroller for embedded applications requiring a high level of integration and low power dissipation. The ARM Cortex-M3 is a next generation core that offers system enhancements such as modernized debug features and a higher level of support block integration.

High speed versions (LPC1769 and LPC1759) operate at up to a 120 MHz CPU frequency. Other versions operate at up to an 100 MHz CPU frequency. The ARM Cortex-M3 CPU incorporates a 3-stage pipeline and uses a Harvard architecture with separate local instruction and data buses as well as a third bus for peripherals. The ARM Cortex-M3 CPU also includes an internal prefetch unit that supports speculative branches.

The peripheral complement of the LPC176x/5x includes up to 512 kB of flash memory, up to 64 kB of data memory, Ethernet MAC, a USB interface that can be configured as either Host, Device, or OTG, 8 channel general purpose DMA controller, 4 UARTs, 2 CAN channels, 2 SSP controllers, SPI interface, 3 I²C interfaces, 2-input plus 2-output I²S interface, 8 channel 12-bit ADC, 10-bit DAC, motor control PWM, Quadrature Encoder interface, 4 general purpose timers, 6-output general purpose PWM, ultra-low power RTC with separate battery supply, and up to 70 general purpose I/O pins.

1.2 Features

Refer to [Section 1.4.1](#) for details of features on specific part numbers.

- ARM Cortex-M3 processor, running at frequencies of up to 120 MHz on high speed versions (LPC1769 and LPC1759), up to 100 MHz on other versions. A Memory Protection Unit (MPU) supporting eight regions is included.
- ARM Cortex-M3 built-in Nested Vectored Interrupt Controller (NVIC).
- Up to 512 kB on-chip flash program memory with In-System Programming (ISP) and In-Application Programming (IAP) capabilities. The combination of an enhanced flash memory accelerator and location of the flash memory on the CPU local code/data bus provides high code performance from flash.
- Up to 64 kB on-chip SRAM includes:
 - Up to 32 kB of SRAM on the CPU with local code/data bus for high-performance CPU access.
 - Up to two 16 kB SRAM blocks with separate access paths for higher throughput. These SRAM blocks may be used for Ethernet, USB, and DMA memory, as well as for general purpose instruction and data storage.
- Eight channel General Purpose DMA controller (GPDMA) on the AHB multilayer matrix that can be used with the SSP, I²S, UART, the Analog-to-Digital and Digital-to-Analog converter peripherals, timer match signals, GPIO, and for memory-to-memory transfers.
- Multilayer AHB matrix interconnect provides a separate bus for each AHB master. AHB masters include the CPU, General Purpose DMA controller, Ethernet MAC, and the USB interface. This interconnect provides communication with no arbitration delays unless two masters attempt to access the same slave at the same time.
- Split APB bus allows for higher throughput with fewer stalls between the CPU and DMA. A single level of write buffering allows the CPU to continue without waiting for completion of APB writes if the APB was not already busy.
- Serial interfaces:
 - Ethernet MAC with RMII interface and dedicated DMA controller.
 - USB 2.0 full-speed controller that can be configured for either device, Host, or OTG operation with an on-chip PHY for device and Host functions and a dedicated DMA controller.
 - Four UARTs with fractional baud rate generation, internal FIFO, IrDA, and DMA support. One UART has modem control I/O and RS-485/EIA-485 support.
 - Two-channel CAN controller.
 - Two SSP controllers with FIFO and multi-protocol capabilities. The SSP interfaces can be used with the GPDMA controller.
 - SPI controller with synchronous, serial, full duplex communication and programmable data length. SPI is included as a legacy peripheral and can be used instead of SSP0.
 - Three enhanced I²C-bus interfaces, one with an open-drain output supporting the full I²C specification and Fast mode plus with data rates of 1Mbit/s, two with standard port pins. Enhancements include multiple address recognition and monitor mode.

- I²S (Inter-IC Sound) interface for digital audio input or output, with fractional rate control. The I²S interface can be used with the GPDMA. The I²S interface supports 3-wire data transmit and receive or 4-wire combined transmit and receive connections, as well as master clock output.
- Other peripherals:
 - 70 (100 pin package) or 52 (80-pin package) General Purpose I/O (GPIO) pins with configurable pull-up/down resistors, open drain mode, and repeater mode. All GPIOs are located on an AHB bus for fast access, and support Cortex-M3 bit-banding. GPIOs can be accessed by the General Purpose DMA Controller. Any pin of ports 0 and 2 can be used to generate an interrupt.
 - 12-bit Analog-to-Digital Converter (ADC) with input multiplexing among eight pins, conversion rates up to 200 kHz, and multiple result registers. The 12-bit ADC can be used with the GPDMA controller.
 - 10-bit Digital-to-Analog Converter (DAC) with dedicated conversion timer and DMA support.
 - Four general purpose timers/counters, with a total of eight capture inputs and ten compare outputs. Each timer block has an external count input. Specific timer events can be selected to generate DMA requests.
 - One motor control PWM with support for three-phase motor control.
 - Quadrature encoder interface that can monitor one external quadrature encoder.
 - One standard PWM/timer block with external count input.
 - Real-Time Clock (RTC) with a separate power domain. The RTC is clocked by a dedicated RTC oscillator. The RTC block includes 20 bytes of battery-powered backup registers, allowing system status to be stored when the rest of the chip is powered off. Battery power can be supplied from a standard 3 V Lithium button cell. The RTC will continue working when the battery voltage drops to as low as 2.1 V. An RTC interrupt can wake up the CPU from any reduced power mode.
 - Watchdog Timer (WDT). The WDT can be clocked from the internal RC oscillator, the RTC oscillator, or the APB clock.
 - Cortex-M3 system tick timer, including an external clock input option.
 - Repetitive interrupt timer provides programmable and repeating timed interrupts.
- Standard JTAG test/debug interface as well as Serial Wire Debug and Serial Wire Trace Port options.
- Emulation trace module supports real-time trace.
- Four reduced power modes: Sleep, Deep-sleep, Power-down, and Deep power-down.
- Single 3.3 V power supply (2.4 V to 3.6 V). Temperature range of -40 °C to 85 °C.
- Four external interrupt inputs configurable as edge/level sensitive. All pins on PORT0 and PORT2 can be used as edge sensitive interrupt sources.
- Non-maskable Interrupt (NMI) input.
- Clock output function that can reflect the main oscillator clock, IRC clock, RTC clock, CPU clock, or the USB clock.
- The Wake-up Interrupt Controller (WIC) allows the CPU to automatically wake up from any priority interrupt that can occur while the clocks are stopped in deep sleep, Power-down, and Deep power-down modes.

- Processor wake-up from Power-down mode via any interrupt able to operate during Power-down mode (includes external interrupts, RTC interrupt, USB activity, Ethernet wake-up interrupt, CAN bus activity, PORT0/2 pin interrupt, and NMI).
- Each peripheral has its own clock divider for further power savings.
- Brownout detect with separate threshold for interrupt and forced reset.
- On-chip Power-On Reset (POR).
- On-chip crystal oscillator with an operating range of 1 MHz to 25 MHz.
- 4 MHz internal RC oscillator trimmed to 1% accuracy that can optionally be used as a system clock.
- An on-chip PLL allows CPU operation up to the maximum CPU rate without the need for a high-frequency crystal. May be run from the main oscillator, the internal RC oscillator, or the RTC oscillator.
- A second, dedicated PLL may be used for the USB interface in order to allow added flexibility for the Main PLL settings.
- Versatile pin function selection feature allows many possibilities for using on-chip peripheral functions.
- Available as LQFP100 (14 mm × 14 mm × 1.4 mm), TFBGA100¹ (9 mm × 9 mm × 0.7 mm), WLCSP100 (5.074 × 5.074 × 0.6 mm) package, and 80-pin LQFP (12 x 12 x 1.4 mm) packages

1.3 Applications

- eMetering
- Lighting
- Industrial networking
- Alarm systems
- White goods
- Motor control

1. LPC1768/65 only.

1.4 Ordering information

Table 1. Ordering information

Type number	Package		
	Name	Description	Version
LPC1769FBD100			
LPC1768FBD100			
LPC1767FBD100	LQFP100	plastic low profile quad flat package; 100 leads; body 14 × 14 × 1.4 mm	SOT407-1
LPC1766FBD100			
LPC1765FBD100			
LPC1764FBD100			
LPC1763FBD100			
LPC1768FET100	TFBGA100	plastic thin fine-pitch ball grid array package; 100 balls; body 9 × 9 × 0.7 mm	SOT926-1
LPC1768UK	WLCSP100	wafer level chip-scale package; 100 balls; 5.074 × 5.074 × 0.6 mm	-
LPC1759FBD80			
LPC1758FBD80			
LPC1756FBD80	LQFP80	plastic low profile quad flat package; 80 leads; body 12 × 12 × 1.4 mm	SOT315-1
LPC1754FBD80			
LPC1752FBD80			
LPC1751FBD80			

1.4.1 Part options summary

Table 2. Ordering options for LPC176x/5x parts

Type number	Max. CPU speed	Flash	Total SRAM	Ethernet	USB	CAN	I ² S	DAC	Package
LPC1769FBD100	120 MHz	512 kB	64 kB	yes	Device/Host/OTG	2	yes	yes	100 pin
LPC1768FBD100	100 MHz	512 kB	64 kB	yes	Device/Host/OTG	2	yes	yes	100 pin
LPC1768FET100	100 MHz	512 kB	64 kB	yes	Device/Host/OTG	2	yes	yes	100 pin
LPC1768UK	100 MHz	512 kB	64 kB	yes	Device/Host/OTG	2	yes	yes	100 pin
LPC1767FBD100	100 MHz	512 kB	64 kB	yes	no	no	yes	yes	100 pin
LPC1766FBD100	100 MHz	256 kB	64 kB	yes	Device/Host/OTG	2	yes	yes	100 pin
LPC1765FBD100	100 MHz	256 kB	64 kB	no	Device/Host/OTG	2	yes	yes	100 pin
LPC1764FBD100	100 MHz	128 kB	32 kB	yes	Device	2	no	no	100 pin
LPC1763FBD100	100 MHz	256 kB	64 kB	no	no	no	yes	yes	100 pin
LPC1759FBD80	120 MHz	512 kB	64 kB	no	Device/Host/OTG	2	yes	yes	80 pin
LPC1758FBD80	100 MHz	512 kB	64 kB	yes	Device/Host/OTG	2	yes	yes	80 pin
LPC1756FBD80	100 MHz	256 kB	32 kB	no	Device/Host/OTG	2	yes	yes	80 pin
LPC1754FBD80	100 MHz	128 kB	32 kB	no	Device/Host/OTG	1	no	yes	80 pin
LPC1752FBD80	100 MHz	64 kB	16 kB	no	Device	1	no	no	80 pin
LPC1751FBD80	100 MHz	32 kB	8 kB	no	Device	1	no	no	80 pin

1.5 Simplified block diagram

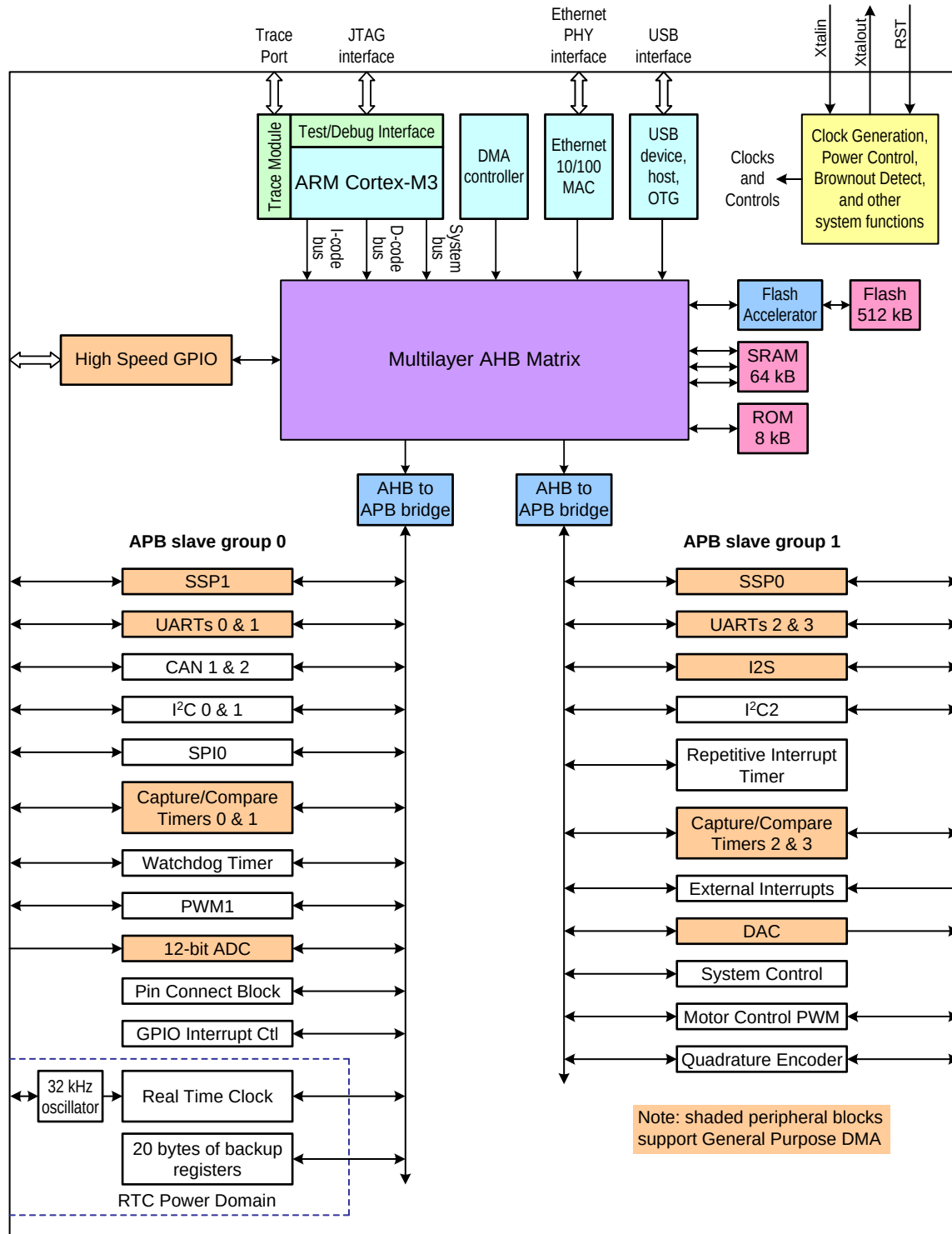


Fig 1. LPC1768 simplified block diagram

1.6 Architectural overview

The ARM Cortex-M3 includes three AHB-Lite buses, one system bus and the I-code and D-code buses which are faster and are used similarly to TCM interfaces: one bus dedicated for instruction fetch (I-code) and one bus for data access (D-code). The use of two core buses allows for simultaneous operations if concurrent operations target different devices.

The LPC176x/5x uses a multi-layer AHB matrix to connect the Cortex-M3 buses and other bus masters to peripherals in a flexible manner that optimizes performance by allowing peripherals on different slaves ports of the matrix to be accessed simultaneously by different bus masters. Details of the multilayer matrix connections are shown in [Figure 2](#).

APB peripherals are connected to the CPU via two APB busses using separate slave ports from the multilayer AHB matrix. This allows for better performance by reducing collisions between the CPU and the DMA controller. The APB bus bridges are configured to buffer writes so that the CPU or DMA controller can write to APB devices without always waiting for APB write completion.

1.7 ARM Cortex-M3 processor

The ARM Cortex-M3 is a general purpose 32-bit microprocessor, which offers high performance and very low power consumption. The Cortex-M3 offers many new features, including a Thumb-2 instruction set, low interrupt latency, hardware divide, interruptible/continuable multiple load and store instructions, automatic state save and restore for interrupts, tightly integrated interrupt controller with Wake-up Interrupt Controller, and multiple core buses capable of simultaneous accesses.

Pipeline techniques are employed so that all parts of the processing and memory systems can operate continuously. Typically, while one instruction is being executed, its successor is being decoded, and a third instruction is being fetched from memory.

The ARM Cortex-M3 processor is described in detail in the Cortex-M3 User Guide that is appended to this manual.

1.7.1 Cortex-M3 Configuration Options

The LPC176x/5x uses the r2p0 version of the Cortex-M3 CPU, which includes a number of configurable options, as noted below.

System options:

- The Nested Vectored Interrupt Controller (NVIC) is included. The NVIC includes the SYSTICK timer.
- The Wake-up Interrupt Controller (WIC) is included. The WIC allows more powerful options for waking up the CPU from reduced power modes.
- A Memory Protection Unit (MPU) is included.
- A ROM Table is included. The ROM Table provides addresses of debug components to external debug systems.

Debug related options:

- A JTAG debug interface is included.
- Serial Wire Debug is included. Serial Wire Debug allows debug operations using only 2 wires, simple trace functions can be added with a third wire.
- The Embedded Trace Macrocell (ETM) is included. The ETM provides instruction trace capabilities.
- The Data Watchpoint and Trace (DWT) unit is included. The DWT allows data address or data value matches to be trace information or trigger other events. The DWT includes 4 comparators and counters for certain internal events.
- An Instrumentation Trace Macrocell (ITM) is included. Software can write to the ITM in order to send messages to the trace port.
- The Trace Port Interface Unit (TPIU) is included. The TPIU encodes and provides trace information to the outside world. This can be on the Serial Wire Viewer pin or the 4-bit parallel trace port.
- A Flash Patch and Breakpoint (FPB) is included. The FPB can generate hardware breakpoints and remap specific addresses in code space to SRAM as a temporary method of altering non-volatile code. The FPB include 2 literal comparators and 6 instruction comparators.

1.8 On-chip flash memory system

The LPC176x/5x contains up to 512 kB of on-chip flash memory. A flash memory accelerator maximizes performance for use with the two fast AHB-Lite buses. This memory may be used for both code and data storage. Programming of the flash memory may be accomplished in several ways. It may be programmed In System via the serial port. The application program may also erase and/or program the flash while the application is running, allowing a great degree of flexibility for data storage field firmware upgrades, etc.

1.9 On-chip Static RAM

The LPC176x/5x contains up to 64 kB of on-chip static RAM memory. Up to 32 kB of SRAM, accessible by the CPU and all three DMA controllers are on a higher-speed bus. Devices containing more than 32 kB SRAM have two additional 16 kB SRAM blocks, each situated on separate slave ports on the AHB multilayer matrix.

This architecture allows the possibility for CPU and DMA accesses to be separated in such a way that there are few or no delays for the bus masters.

1.10 Block diagram

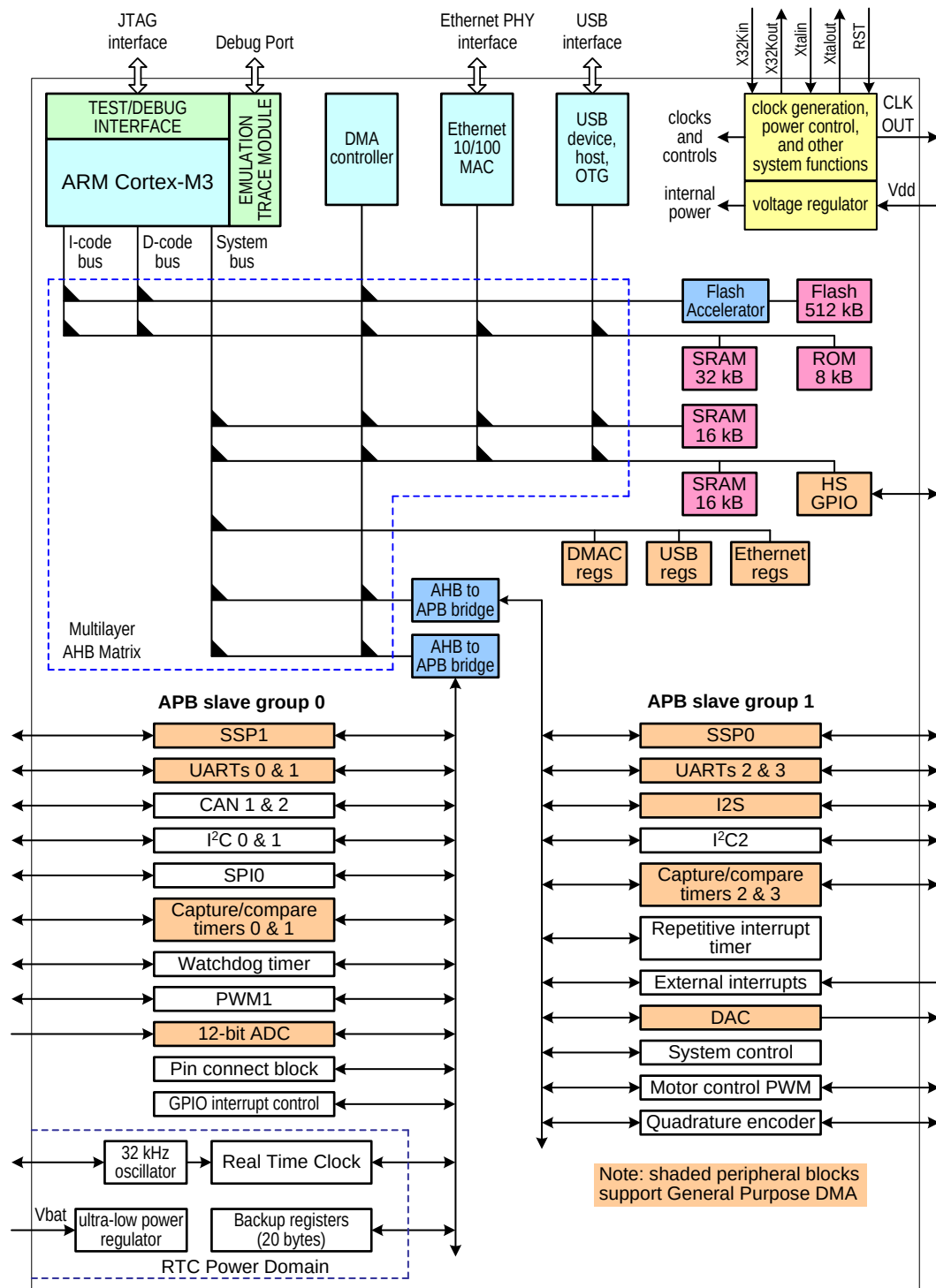


Fig 2. LPC1768 block diagram, CPU and buses